

# Advanced Calculus HW 7

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## Question 1

We know that the tangent plane to  $F$  at  $(x_0, y_0, z_0)$  is given by

$$F(x_0, y_0, z_0) + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

And its normal line is given by  $\vec{l}(t) = (x_0, y_0, z_0) + t(\nabla f(x_0, y_0, z_0))$

(a)

$$F(x, y, z) = z - 2 - \frac{1}{2} \ln(x^2 + y^2)$$

$$F(-1, 0, 2) = 0$$

$$\nabla F(x, y, z) = \left( \frac{-x}{x^2 + y^2}, \frac{-y}{x^2 + y^2}, 1 \right)$$

$$\nabla F(-1, 0, 2) = (1, 0, 1)$$

Then the tangent plane is given by

$$(x + 1) + (z - 2) = 0$$

$$x + z - 1 = 0$$

And the normal line is

$$\vec{l}(t) = (-1, 0, 2) + t(1, 0, 1)$$

(b)

$$F(x, y, z) = z - \frac{1}{2} x^7 y^{-2} - 1$$

$$F(2, 4, 3) = 0$$

$$\nabla F(x, y, z) = \left( -\frac{7}{2} x^6 y^{-2}, x^7 y^{-3}, 1 \right)$$

$$\nabla F(2, 4, 3) = (-14, 2, 1)$$

Then the tangent plane is given by

$$-14(x - 2) + 2(y - 4) + z - 3 = 0$$

$$-14x + 2y + z + 17 = 0$$

And the normal line is

$$\vec{l}(t) = (2, 4, 3) + t(-14, 2, 1)$$

## Question 2

(a)

$$w = f(\rho) = f(\sqrt{(x^2 + y^2 + z^2)})$$

$$\frac{\partial w}{\partial x} = \frac{dw}{d\rho} \frac{\partial \rho}{\partial x}$$

with  $\frac{\partial \rho}{\partial x} = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{\rho}$ , since  $\rho$  is similar across  $x, y$  and  $z$ , their partial derivatives are similar  
Then

$$\begin{aligned} \left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2 &= \left(\frac{df}{d\rho} \frac{x}{\rho}\right)^2 + \left(\frac{df}{d\rho} \frac{y}{\rho}\right)^2 + \left(\frac{df}{d\rho} \frac{z}{\rho}\right)^2 \\ &= \left(\frac{df}{d\rho}\right)^2 \left(\frac{x^2}{\rho^2} + \frac{y^2}{\rho^2} + \frac{z^2}{\rho^2}\right) \\ &= \left(\frac{df}{d\rho}\right)^2 \left(\frac{1}{\rho^2}\right) (x^2 + y^2 + z^2) \\ &= \left(\frac{df}{d\rho}\right)^2 \left(\frac{\rho^2}{\rho^2}\right) \\ &= \left(\frac{df}{d\rho}\right)^2 \end{aligned}$$

Also

$$\frac{dw}{d\rho} = \frac{df}{d\rho}$$

Then

$$\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2 = \left(\frac{dw}{d\rho}\right)^2$$

(b)

$$w = f(x, y, g(x, y))$$

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial g} \frac{\partial g}{\partial x}$$

$$\frac{\partial w}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial g} \frac{\partial g}{\partial y}$$

(c)

$$w = f(x, y, z)$$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$

$$= \left(\frac{6x}{3x^2 - 2y + 4z^3}\right) \left(\frac{1}{2\sqrt{t}}\right) + \left(-\frac{2}{3x^2 - 2y + 4z^3}\right) \left(\frac{2}{3}t^{-\frac{1}{3}}\right) + \left(\frac{12z^2}{3x^2 - 2y + 4z^3}\right) \left(-\frac{1}{t^2}\right)$$

$$= \left(-\frac{2}{3x^2 - 2y + 4z^3}\right) \left(-\frac{3x}{2\sqrt{t}} + \frac{2}{3t^{\frac{1}{3}}} + \frac{6z^2}{t^2}\right)$$

## Question 3

(a)

First normalise  $\vec{u}$

$$\hat{u} = \left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, -\frac{3}{\sqrt{14}}\right)$$

$$\nabla f(x, y, z) = \left(-\frac{y}{(x^2 + z^2)^2}, \frac{1}{x + z}, -\frac{y}{(x^2 + z^2)^2}\right)$$

Then the directional derivative is given by  $\hat{u} \cdot \nabla f(x_0, y_0, z_0)$

$$\left(-\frac{y_0}{\sqrt{14}(x_0^2 + z_0^2)^2}, \frac{2}{\sqrt{14}(x_0 + z_0)}, \frac{3y_0}{\sqrt{14}(x_0^2 + z_0^2)^2}\right)$$

(b)

$$f(x, y, z) = y^2 z \tan^3 x$$

$$f_x(x, y, z) = 3y^2 z \tan^2 x \sec^2 x$$

$$f_y(x, y, z) = 2yz \tan^3 x$$

$$f_z(x, y, z) = y^2 \tan^3 x$$

$$\Rightarrow \nabla f(x, y, z) = (3y^2 z \tan^2 x \sec^2 x, 2yz \tan^3 x, y^2 \tan^3 x)$$

$$\Rightarrow \nabla f(\pi/4, -2, 1) = (24, -4, 4)$$